F1 RACE INTELLIGENCE

Tableau 2026 Project Report & Implementation Guide

Dark motorsport theme · 9-scene data story · 5-sheet relational data model

1. Project overview

This project analyzes the supplied Formula 1 Race Intelligence workbook across race results, championship standings, circuits, lap intelligence, and qualifying results. The intended dashboard uses a black background, red accents, white text, four KPI cards, three controls, and eight analytical views. The narrative progresses from overall scale to geography, reliability, driver and team performance, championship trends, tyre usage, country participation, and podium achievements.

Validation note: the supplied file contains 13,748 data rows across the five source sheets. The race_results sheet contains 6,464 rows and 42 columns. These measured values are used instead of approximate counts in the brief.

2. Dataset inventory

Sheet	Rows	Columns
circuits	29	9
race_results	6,464	42
championship_standings	311	10
lap_intelligence	3,814	14
qualifying_results	3,130	16
TOTAL	13,748	Across source sheets

3. Data model and relationships

• Connect the Excel workbook through Tableau’s Microsoft Excel connector and use the five named sheets.

• Use logical relationships rather than flattening all sheets into one physical join. Keep championship_standings independent for championship visuals to avoid multiplying race-result measures.

• Relate race_id to race-level results, lap intelligence, and qualifying records where valid; use circuit_id for circuit metadata. Confirm key uniqueness/cardinality before building relationships.

• Use an extract for responsive dashboard filtering if supported by the target Tableau installation.

• Preserve all DNF rows. The source contains 396 missing finish_position values and 396 missing gap_to_leader_s values; do not discard these records from reliability analysis. Exclude null gap values only from gap-based aggregations.

• Use race_date as a date; season and positions as numeric dimensions; points and counts as measures; driver, team, country, tyre compound, and DNF reason as dimensions.

4. Dashboard design and filters

Canvas: black/charcoal background (#080808 to #151515), red accent (#E10600), white titles, muted gray gridlines. Top band: F1-inspired title and racing-car image. KPI row: Competing Teams, Seasons in Play, Race Outcomes Logged, Circuits Showcased. Controls: DNF Reason, Year of Race Date, Top N Parameter. Lower grid: row 1 map + reliability stacked bar; row 2 driver leaderboard + team ranking; row 3 championship grouped bar, tyre pie, country treemap/highlight table, podium bubbles.

DNF Reason dropdown

Normalize the source reasons to the requested choices: Engine → Engine failure; Brake failure → Brake failure; Collision → Collision; Mechanical → Mechanical; Hydraulics and Gearbox → Other. Keep non-DNF records as not applicable rather than assigning a false DNF reason.

Year of Race Date

Create YEAR([race_date]) as a year dimension and apply it to relevant race-result worksheets. For championship_standings, use season and synchronize the year selection through a season field or carefully designed parameter/action.

Top N Parameter

Create an integer parameter named Top N Parameter, default 10, minimum 1, maximum 19 (or the actual distinct driver count). Use a Boolean calculated field to keep ranks up to the selected parameter and apply it only to the driver leaderboard.

5. Nine-scene data story

Scene	Title	Chart	Metric / implementation
1	Global overview	4 KPI cards	Teams, seasons, result rows, circuits
2	Global driver distribution across Grand Prix nations	Geographic map	Country-level driver counts; not exact circuit coordinates
3	Driver reliability analysis	Stacked bar	DNF counts by driver and normalized reason
4	Driver performance leaderboard	Vertical bar	Points; Top N parameter
5	Team performance ranking	Horizontal bar	Team points or clearly labeled metric
6	Team championship performance analysis	Grouped vertical bar	Season/team championship points
7	Tyre compound usage distribution	Pie	Starting compound or lap-level compound, clearly labeled
8	Country-wise race participation	Treemap / highlight table	Race-result entries or distinct races by country
9	Podium achievements by driver and team	Packed bubble	Count finish_position 1–3; color/group by team

6. Suggested minimal calculations

Name	Formula	Notes
DNF Count	SUM([dnf])	dnf is a 0/1 flag
Year of Race Date	YEAR([race_date])	Year selector
Keep Top N Drivers	RANK(SUM([points])) <= [Top N Parameter]	Compute using Driver; sort points descending
Podium Count	SUM(IIF([finish_position] <= 3 AND [finish_position] >= 1, 1, 0))	Count actual podium rows
Points Cumulative (optional)	RUNNING_SUM(SUM([points]))	Partition by driver and season when needed

7. Tableau build and validation checklist

- Build a worksheet for each analytical view and a KPI group for the four summary metrics.
- Use clear titles, sort bars descending, and include informative tooltips and aggregation labels.
- Keep Top N limited to the driver leaderboard; configure DNF and year filters on the intended sheets.
- Test all filter combinations for blank views, duplicated totals, and unexpected null categories.
- Set dashboard size to Automatic or a fixed desktop size such as 1440 × 900, then test a smaller layout.
- Open the Excel workbook in Tableau Desktop and save the verified result as a Tableau Packaged Workbook (.twbx).

8. Submission files and limitation

This package contains the Tableau-ready Excel data workbook, interactive HTML dashboard preview, DOCX report/build guide, PDF report, README, and dataset validation CSV. The HTML preview is a companion visualization, not a native Tableau workbook. A .twb/.twbx has not been generated or validated in Tableau 2026 in this environment. Use the supplied Excel file and guide to create/open the native workbook in Tableau Desktop, then save it as a packaged workbook for final Tableau submission.